Eisha Ashraf

Physics Graduate | Medical Physics & Biomedical Engineering | AI in Healthcare

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ACHIEVEMENTS & HIGHLIGHTS

- BS Physics graduate with a 3.70/4.00 CGPA and project experience spanning medical imaging, biomedical applications, AI, and materials science.
- Developed healthcare-focused AI projects and participated in international hackathons using Python, Streamlit, GitHub, and AI tools.
- IELTS Academic: 6.5 overall. Duolingo English Test: 115.
- Completed a Final Year Project on Water Treatment Through Magnetic Materials using Nickel Ferrite (NiFe■O■) nanoparticles for fluoride removal.

EDUCATION

B.S. in Physics — COMSATS University Islamabad, Pakistan | Completed 2026

CGPA: 3.70/4.00 | Relevant areas: Medical Physics, Modern Physics, Electromagnetism, Quantum Physics, Nuclear Physics, Materials Science, Computational Physics, Data Analysis and Scientific Computing.

English: IELTS Academic 6.5 | Duolingo English Test 115

ACADEMIC & RESEARCH PROJECTS

Artificial Intelligence in Medical Imaging for Early Cancer Detection

- Investigated AI and Machine Learning applications for early cancer detection through medical imaging and computer-assisted diagnosis.
- Explored abnormal-pattern recognition, clinical decision support, dataset limitations, model reliability, and validation challenges.

Comparison of MRI, CT, PET, and Ultrasound for Cancer Diagnosis

- Compared imaging principles, clinical applications, advantages, limitations, resolution, and radiation exposure across four major modalities.
- Analyzed suitability of different modalities for cancer-detection scenarios.

Magnetic Nanoparticles in Cancer Therapy and Drug Delivery

- Investigated magnetic drug targeting, MRI-related applications, targeted drug delivery, and magnetic hyperthermia.
- Examined biomedical potential, challenges, biocompatibility considerations, and future applications.

FINAL YEAR PROJECT

Water Treatment Through Magnetic Materials — Nickel Ferrite (NiFe■O■) Porous Magnetic Nanoparticles for Fluoride Removal

- Synthesized Nickel Ferrite magnetic nanoparticles using a hydrothermal method and investigated their use as adsorbents for fluoride removal.
- Studied adsorbent dosage, pH, contact time, temperature, surface area, and initial pollutant concentration.
- Investigated magnetic separation for recovery and potential reusability.
- Worked with XRD, FTIR, SEM, BET, and VSM characterization concepts and experimental data interpretation.
- Evaluated fluoride removal performance against the WHO guideline value of 1.5 mg/L.

INTERNATIONAL HACKATHON PROJECTS

NeuroRecover AI — Hack for Humanity | Summer 2026

- Developed an AI-powered healthcare project focused on concussion recovery support and tracking.

- Implemented using Python and Streamlit; completed GitHub, deployment, documentation, demo, and submission workflow.

AI Health Assistant

- Developed a Streamlit healthcare application integrating a BMI Calculator, AI Symptom Checker, and Medical Imaging Guide.

OncoAssist AI

- Developed a healthcare-focused AI project exploring technology-assisted support for cancer-related applications.

TECHNICAL SKILLS

Programming: Python, MATLAB, Git, GitHub, Streamlit, Google Colab, Visual Studio Code

AI & Data: Artificial Intelligence, Machine Learning, Medical Imaging AI, Data Analysis, NumPy, pandas, Matplotlib, Excel

Biomedical & Physics: Medical Physics, Medical Imaging, Biomedical Engineering, Nanotechnology, Magnetic Nanoparticles, Materials Science, Cancer Research

Research: Scientific Research, Scientific Writing, Research Methods, Data Interpretation, Technical Documentation, Scientific Presentation

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Introduction to Modern AI — Cisco Networking Academy
- Python Essentials 1 & 2 — Cisco Networking Academy
- Frontiers of Biomedical Engineering — Yale University
- Artificial Intelligence learning — University of Helsinki
- AI / technical learning — IBM SkillsBuild
- Research-focused learning including Writing Skills, Ethics, Research Design, Research Data Management, Finding the Right Journal, Fundamentals of Manuscript Preparation, and Ensuring Visibility — Elsevier
- Additional relevant technical and professional learning through Coursera and other online programs.

RESEARCH INTERESTS

Medical Physics · Biomedical Engineering · Medical Imaging · AI in Healthcare · AI-assisted Cancer Detection · Medical Image Analysis · Nanotechnology · Magnetic Nanoparticles · Cancer Therapy · Drug Delivery · Computational Physics · Biomedical Materials